MATH 345 Skeleton Notes
Unit 2: Anatomy of a linear map
Section 2.3: Subspaces

Definition of subspaces

The outputs a matrix can "reach" form a subspace. The directions a matrix "forgets" also form a subspace. The definitions below make this precise. In particular, the closure rules are not arbitrary. If two output vectors are reachable, then any linear combination of them is reachable. If two input changes are forgotten by a matrix, then any linear combination of those changes is also forgotten.

We are already familiar with geometric spaces living inside bigger ones, like lines lying in the plane, or planes lying in 3-space. We are now going to formalize this notion algebraically. Recall the definition of the set $\mathbb{R}^n$ in Vector.

Definition: Subspaces A set U of vectors in $\mathbb{R}^n$ is called a $\boxed{\text{subspace}}$ of $\mathbb{R}^n$ if it satisfies the following properties:

- The zero vector (written as $\mathbf{0}$) is in U.
- If $\mathbf{x}$ is in U and y is in U, then $\mathbf{x} + \mathbf{y}$ is in U. This means U is *closed under addition*.
- If $\mathbf{x}$ is in U, then $a\mathbf{x}$ is in U for every scalar a. This means U is *closed under scalar multiplication*.

Example The whole set $\mathbb{R}^n$ is a subspace of itself.

Example The set containing only the zero vector, i.e., the set $\{\mathbf{0}\}$, is a subspace of $\mathbb{R}^n$ (called the trivial subspace or zero subspace).

Activity

Verify that $\{\mathbf{0}\}$ is a subspace of $\mathbb{R}^n$.

Definition: Proper Subspaces Any subspace of $\mathbb{R}^n$ other than $\mathbb{R}^n$ or $\{0\}$ is a proper subspace of $\mathbb{R}^n$.

Example Planes and lines through the origin in $\mathbb{R}^3$ are all subspaces of $\mathbb{R}^3$.

Activity

Verify that planes through the origin in $\mathbb{R}^3$ are all proper subspaces of $\mathbb{R}^3$.

Activity

Verify that lines through the origin in $\mathbb{R}^3$ are subspaces of $\mathbb{R}^3$.

Note Lines and planes which do not pass through the origin in $\mathbb{R}^3$ are *not* subspaces of $\mathbb{R}^3$. Nonetheless, if Π is a plane, and we consider the set of displacement vectors

$$\Delta\Pi = \{\mathbf{v}_1 - \mathbf{v}_0 \ : \ \mathbf{v}_0, \mathbf{v}_1 \in \Pi\},$$

then $\Delta\Pi$ forms a plane passing through the origin, and so is a subspace of $\mathbb{R}^3$. A similar result holds for sets of displacement vectors given by a line not passing through the origin.

Activity

Show that the set

$$U = \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \ : \ x_1 x_2 = 0 \right\}$$

is *not* a subspace of $\mathbb{R}^2$.

Activity

Let W be the set of all vectors in $\mathbb{R}^3$ of the form

$$\begin{bmatrix} r \\ s \\ r+2s \end{bmatrix},$$

where $r, s \in \mathbb{R}$. Is W a proper subspace of $\mathbb{R}^3$?

Note To show a set *is* a subspace, you tend to need to parameterize all elements of the set in order to show they all satisfy some property (closure under addition or scalar multiplication), and so your argument should use variables. To show a set *is not* a subspace, you need to find one or more counterexamples that show that some property does not hold for all elements; a single explicit counterexample usually suffices.

Null space and image space

The following important examples are representative of two canonical ways of describing a subspace.

From here on, homogeneous equations play a special role. To solve $A\mathbf{x} = \mathbf{0}$ by row reduction, we could write the augmented matrix $[A \mid \mathbf{0}]$. Since the right-hand side stays zero throughout the row operations, the coefficient matrix A carries the essential information.

Definition: Null Space and Image Space Let A be an $m \times n$ matrix.

The $\boxed{\text{null space}}$ of A, denoted $\operatorname{null}(A)$, is the set of vectors $\mathbf{x}$ in $\mathbb{R}^n$ which are solutions to the equation $A\mathbf{x} = \mathbf{0}$, i.e.,

$$\operatorname{null}(A) = \{\mathbf{x} \in \mathbb{R}^n \,:\, A\mathbf{x} = \mathbf{0}\}.$$

The $\boxed{\text{image space}}$ of A, denoted $\operatorname{im}(A)$, is the set of vectors $\mathbf{y}$ in $\mathbb{R}^m$ such that $A\mathbf{x} = \mathbf{y}$ has a solution, i.e.,

$$\operatorname{im}(A) = \{A\mathbf{x} \,:\, \mathbf{x} \in \mathbb{R}^n\}.$$

Note The null space lives in the input space. It records input directions that A sends to $\mathbf{0}$.
The image space lives in the output space. It records outputs that A can reach.

- **Object:** $\operatorname{null}(A)$
  Lives in: $\mathbb{R}^n$
  How to read it: input directions A forgets

- **Object:** $\operatorname{im}(A)$
  Lives in: $\mathbb{R}^m$
  How to read it: outputs A can reach

Activity: Revisiting the height map with subspace language

Return to A map that forgets height. Let

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Describe the output subspace and the subspace of input directions that P forgets.

Theorem For any $m \times n$ matrix, the null space of A is a subspace of $\mathbb{R}^n$, and the image space of A is a subspace of $\mathbb{R}^m$.

Proof We start by showing the null space of A is a subspace. Recall the properties discussed in Properties of matrix-vector products:
Contains the zero vector $\mathbf{0} \in \operatorname{null}(A)$ because $\mathbf{A0} = \mathbf{0}$.
Closure under addition
Suppose $\mathbf{v} \in \operatorname{null}(A)$ and $\mathbf{w} \in \operatorname{null}(A)$. Then $A\mathbf{v} = \mathbf{0}$ and $A\mathbf{w} = \mathbf{0}$. So

$$A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w} = \mathbf{0} + \mathbf{0} = \mathbf{0}.$$

Thus $\mathbf{v} + \mathbf{w} \in \operatorname{null}(A)$.
Closure under scalar multiplication
If $\mathbf{v} \in \operatorname{null}(A)$, and a is a scalar, then

$$A(a\mathbf{v}) = a(A\mathbf{v}) = a\mathbf{0} = \mathbf{0}.$$

Thus $a\mathbf{v} \in \operatorname{null}(A)$.
So $\operatorname{null}(A)$ is a subspace of $\mathbb{R}^n$.
Next, we show that $\operatorname{im}(A)$ is a subspace.
Contains the zero vector $\mathbf{0} \in \operatorname{im}(A)$ because $\mathbf{A0} = \mathbf{0}$.
Closure under addition
Suppose $\mathbf{v} \in \operatorname{im}(A)$ and $\mathbf{w} \in \operatorname{im}(A)$. Then the equations $A\mathbf{x} = \mathbf{v}$ and $A\mathbf{x} = \mathbf{w}$ both have solutions, i.e., there exists $\mathbf{x}_1$ and $\mathbf{x}_2$ such that $A\mathbf{x}_1 = \mathbf{v}$ and $A\mathbf{x}_2 = \mathbf{w}$. Then

$$A(\mathbf{x}_1 + \mathbf{x}_2) = A\mathbf{x}_1 + A\mathbf{x}_2 = \mathbf{v} + \mathbf{w},$$

so the equation $A\mathbf{x} = \mathbf{v} + \mathbf{w}$ has a solution, and so $\mathbf{v} + \mathbf{w} \in \operatorname{im}(A)$.

Closure under scalar multiplication
If $\mathbf{v} \in \operatorname{im}(A)$, and a is a scalar, then we may write $\mathbf{v} = A\mathbf{x}_1$ for some $\mathbf{x}_1$. But then

$$A(a\mathbf{x}_1) = a(A\mathbf{x}_1) = a\mathbf{v},$$

so the equation $A\mathbf{x} = a\mathbf{v}$ has a solution, so that $a\mathbf{v} \in \operatorname{im}(A)$.
Thus $\operatorname{im}(A)$ is a subspace of $\mathbb{R}^m$.

Warning The solution set of $A\mathbf{x} = \mathbf{0}$ is a subspace. The solution set of $A\mathbf{x} = \mathbf{b}$ for $\mathbf{b} \neq \mathbf{0}$ is usually not a subspace; when it is nonempty, it is a shifted copy of $\operatorname{null}(A)$.

Spanning Sets

Even though subspaces of $\mathbb{R}^n$ can have infinitely many vectors, it is possible to represent them with a finite amount of data. Recall that a *linear combination* is a sum of scalar multiples of vectors.

Activity

Show that there are two vectors (describe them explicitly), so that every vector in the set W discussed in the activity can be written as the linear combination of those two vectors.

Definition: The Span of a Set of Vectors The set of all linear combinations of a set of vectors $\mathbf{x}_1, \ldots, \mathbf{x}_k$ is called the span of the vectors $\mathbf{x}_1, \ldots, \mathbf{x}_k$, and is denoted

$$\operatorname{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\} = \{t_1\mathbf{x}_1 + \cdots + t_k\mathbf{x}_k \,:\, t_1, \dots, t_k \in \mathbb{R}\}.$$

If $V = \operatorname{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$, we say that V is $\boxed{\text{spanned}}$ by the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$, and that the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ $\boxed{\text{span}}$ the space V.

For an image space, spanning vectors come from the columns of the matrix. If

$$A = [\mathbf{a}_1 \; \mathbf{a}_2 \; \cdots \; \mathbf{a}_n]$$

and

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$$

then

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

Thus every output $A\mathbf{x}$ is a linear combination of the columns of A.

Theorem Let $U = \operatorname{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ be a set in $\mathbb{R}^n$. Then

- U is a *subspace* of $\mathbb{R}^n$ containing each of the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$.
- If W is a subspace of $\mathbb{R}^n$ and each of the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ is in W, then U is a *subset* of W, i.e., $U \subseteq W$.

Proof Define an $n \times k$ matrix A, with the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ as columns. Then $U = \operatorname{im}(A)$, and is thus a subspace of $\mathbb{R}^n$. Since $\mathbf{x}_i = A\mathbf{e}_i$ for $1 \le i \le k$, the set U contains each of the vectors $\mathbf{x}_i$.
Conversely, suppose W is a subspace of $\mathbb{R}^n$ and $\mathbf{x}_1, \dots, \mathbf{x}_k \in W$. Since W is closed under addition and scalar multiplication, all of the linear combinations of $\mathbf{x}_1, \dots, \mathbf{x}_k$ are elements of W, and so $U \subseteq W$.

Example Recall the standard basis $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ of $\mathbb{R}^n$ from Standard basis. Note that if

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \cdots \\ x_n \end{bmatrix},$$

then

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n.$$

This means that $\mathbb{R}^n = \operatorname{span}\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$.

Activity

In $\mathbb{R}^3$, consider the two vectors

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}.$$

Determine if the vector

$$\mathbf{w} = \begin{bmatrix} 1 \\ 5 \\ -7 \end{bmatrix}$$

belongs to $\operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\}$.

Activity

Find a spanning set for the null space of the matrix

$$A = \begin{bmatrix} 1 & -2 & -4 & 3 \\ 0 & -1 & -3 & 2 \\ 2 & 1 & 7 & -4 \end{bmatrix}.$$

Activity: A spanning set for an image space

Find a spanning set for the image space of the matrix

$$A = \begin{bmatrix} 1 & -2 & -4 & 3 \\ 0 & -1 & -3 & 2 \\ 2 & 1 & 7 & -4 \end{bmatrix}.$$

To summarize: basic solutions of $A\mathbf{x} = \mathbf{0}$ form a spanning set for $\operatorname{null}(A)$, and the columns of A form a spanning set for $\operatorname{im}(A)$.
The next section studies when a spanning set has redundant vectors and how to remove them.